

PIPING ARRANGEMENT EDIT

Arrangement Edit (E3D legacy)

Early E3D arrangement scripts (01–103) — line-no delete/stretch, dimension, tag layer change. Migrating to the new ACE_E3D.

Key Purpose

Why this program

1. Auto-clean E3D-exported GA drawings

– Runs the old 01-07 pipeline: line-no delete, stretch, dimension, tag layer change, mask

→ **Effect:** Raw AVEVA E3D DXF becomes a tidy arrangement drawing in one click, replacing hours of manual line-note and layer edits

2. Consistent elevation & leader conventions

– Force BOP/COP/TOP elevation mode and EL/FL/GL prefix; keep User-Edited or Default leaders

→ **Effect:** Every sheet reads the same way, so reviewers trust the numbers and re-work drops

3. Batch many sheets in parallel

– Multi-file select processed with a process pool and in-memory DXF chaining

→ **Effect:** A whole drawing set is regenerated at once, using all CPU cores instead of one file at a time

Required inputs : One or more arrangement DXF files exported from AVEVA E3D

Main Window – Item Guide

The screenshot shows the 'Piping GA Drawing Option for E3D(old)' window. It features three tabs at the top: 'General' (selected), 'Annotation', and 'Dimension'. The 'General' tab contains the following elements:

- Select Dxf Arrangement Dwg From AVEVA E3D :** A list box (2) showing no files selected. Below it are buttons: 'Select' (1), 'Remove Selected' (3), 'Clear All', and 'Open'.
- Elevation Expression:** Two groups of buttons. The first group has 'BOP' (4) and 'TOP'. The second group has 'EL.' (5) and 'GL.'.
- Nozzle Table Create:** A section with three rows:
 - 'Red chained Leader :': Two buttons, 'User Edited Leader' (6) and 'Default Leader'.
 - 'Inst Valve measure Tag included text delete :': A text input field (7) containing '1'.
 - 'Dim merge base (Default Leader, default 100) :': A text input field (8) containing '1'.
- All Dwg Text Font :** A dropdown menu (9) set to 'Arial'.
- Footer:** Four buttons: 'Apply' (10), 'Save', 'Load' (11), and 'Download'.

1. General / Annotation / Dimension tabs
2. Selected DXF arrangement list from E3D
3. Select / Remove Selected / Clear All / Open
4. Elevation Expression: BOP / COP / TOP
5. Elevation prefix: EL. / FL. / GL.
6. Red chained Leader: User Edited / Default
7. Inst Valve measure Tag delete text
8. Dim merge base (Default Leader, default 100)
9. All Dwg Text Font
10. Apply - run the 01-07 pipeline
11. Save / Load / Download options

Processing Steps

What 'Apply' runs, in order

1. Copy each selected DXF into a temp workspace and chain steps in memory
2. 00 elev: recalculate elevation numbers into the chosen BOP/COP/TOP mode
3. 01 LDEL: delete line-note text (User-Edited vs Default leader branch)
4. 02 STRETCH: stretch leaders and notes into place
5. 03 DIM: add dimensions and merge duplicate line dims by merge base
6. 04 LAYER: reclassify tag layers and delete Inst-Valve tags matching text
7. 05 TAGARR: arrange the tag annotations
8. 06 CTEXT: convert center text, snap line-no and elevation to steel line
9. 07 MASK: apply background mask behind text
10. Save the final result as <name>_ALL.DXF in the source folder

Overview

What it does

1. Early E3D arrangement scripts (01–103) — line-no delete/stretch
2. Dimension
3. Tag layer change
4. Migrating to the new ACE_E3D

Category: Piping Arrangement Edit

Folder: PIPING_ARRANGEMENT_EDIT/E3D

Main scripts: (component scripts)

Scripts / Status: 11 scripts · In Development

Tags: E3D · legacy

Notes & Information

- Part of the in-house Plant Engineering automation suite — category: Piping Arrangement Edit.
- Runs offline as a pure-Python / Tkinter tool; portable across PCs via OneDrive sync (needs Python + packages).
- A set of component scripts — run the relevant step from the launcher / script list.
- Current status: In Development — behavior/outputs may still change.
- Output samples and the program window above reflect the current build.

Thank You

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